

Paul Abrams

Toronto / Kingston, Ontario

Senior Full-Stack Developer · TypeScript / React · Node.js / NestJS · .NET / C# · PostgreSQL & SQL · Cloud, Docker & CI/CD · 15+ Years Shipping Web Applications

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Professional Summary

Senior full-stack developer with **15+ years across the full SDLC**, shipping **web-based B2B and B2C applications** from front-end UI through service back ends and databases. Front end in **React + TypeScript** (also Next.js, Blazor, Knockout.js MVVM); back end in **.NET / C# and Node.js / NestJS / Express**, designing **REST APIs**, data-access layers, and **n-tier and cloud-native microservices** architectures. I've shipped products **solo** (a Dockerized multiplayer game on Node + PostgreSQL; a React training app on a public REST API) and delivered **at enterprise scale** (an n-tier ASP.NET platform serving **5,000+ users** with 24/7 high availability across load-balanced, geographically distributed servers). Strong **SQL** engineering across **SQL Server, Azure SQL, and PostgreSQL** — schema design, stored procedures, and execution-plan tuning — with working **MongoDB** exposure. Comfortable across **cloud and DevOps: Microsoft Azure** (Functions, App Service, API Management, Key Vault, Entra), **Docker / Kubernetes**, and **CI/CD** (Azure DevOps, GitHub Actions). **Recurring pattern:** self-initiated automation that pays back substantial labour cost, and **modernization by deletion** — replacing over-engineered systems with simpler, well-tested ones. **Professional Scrum Master I**. Active open-source contributor — **60+ public PRs** to outside repos including Microsoft's `vsts-team-calendar`, `tfs-cli`, the TypeScript Website, and the **Microsoft Authentication Library .NET** docs.

Professional Experience

Abrams Software Kingston / Toronto, Ontario (Remote)

August 2025 – Present

Full-Stack Developer · Independent

- **TWNR** — architected and shipped a public-facing B2C multiplayer game as a full-stack web application, solo. Dockerized TypeScript npm monorepo: **Node.js (Express) + WebSocket** server backed by **PostgreSQL**, terminal-style xterm.js client served by Vite, per-player sign-up with **JWT auth**, and a **JSON-Schema-generated REST API contract** that doubles as live documentation. Concurrency-correct game state via PostgreSQL transactions (`SELECT ... FOR UPDATE`). Live at twnr.paulabrams.ca.
- **Chess Attack Visualization** — **React 19 + TypeScript** training app (Vite, React Testing Library + Vitest / Jest) consuming the **Lichess.org REST API**, with a **NestJS** (TypeScript decorator framework analogous to **Spring Boot**) auth proxy implementing the **OAuth2 PKCE** flow. Diagnosed a missing feature in `chess.js`, opened [upstream PR #572](#), and published a fork on npm. **Portfolio site** runs as **Astro on GitHub Pages with GitHub Actions** CI/CD.
- **Next.js + React 19 + TypeScript** open-source contributions to physicshub.github.io, a public physics-simulation site — led a **JavaScript → TypeScript migration** of the core engine, introduced **branded nominal types** with TDD coverage, and fixed latent energy-conservation and boundary-collision bugs.
- **LLM code evaluation** for [DataAnnotation.tech](https://dataannotation.tech) and Stellar.ai — prompt fixtures and grading rubrics across **TypeScript, React, Node / NestJS, .NET / C#, Python, Go, Rust, SQL, and MongoDB**, including **security audits with CWE / CVSS-graded inventories**. **60+ public PRs** to outside repos including the **Microsoft Authentication Library .NET** docs and `canada-ca/platform-security`.

Employment and Social Development, Canada Gatineau, QC

December 2021 – July 2025

Senior Software Engineer — Cloud-Native Microservices (BDM Project)

- Enhanced and maintained a **cloud-native microservices** enterprise onboarding application — **.NET / C# back end, Blazor front end, Azure SQL, Azure Functions, GraphQL**. **Designed and consumed REST APIs between services**; integrated enterprise IAM via **Azure AD / Entra (OAuth**

- 2.0, OpenID Connect); published and governed APIs through **Azure API Management** with authentication, throttling, and transformation policies.
- Designed and shipped **custom web extensions in React, TypeScript, and Node.js** — a build/deploy analytics widget rendering success-rate charts via the ADO Builds REST API, and a people-by-project reporter that replaced an unusably slow query with an in-memory hashmap. Responsive UI, custom styling, **WCAG / AODA**-aligned interactions.
- Migrated the application's downstream identity from rotating secrets (Key Vault + ADO PATs) to **Azure Managed Identities** — eliminating the secret-rotation incident class. Operated **CI/CD pipelines** in Azure DevOps for build, test, **Docker** containerization, **Kubernetes** deployment, and **Azure Data Factory** (ETL) orchestration; daily Git workflow with feature branches and PR-gated CI.
- Ran weekly **demos and design walkthroughs**, mentoring junior developers on microservice integration, app registration, and custom extensions.

Employment and Social Development, Canada Gatineau, QC August 2019 – November 2021

Development Lead — Internal Web Application & CI/CD Automation

- Co-architected and developed an **internal C# / .NET web application** with a web UI for authoring structured scenarios; designed the **SQL Server** schema and **co-designed the REST API** for bidirectional integration with **Azure DevOps pipelines** (versioned script-fetch and structured results-push endpoints). Integrated with the departmental headless CMS.
- Engineered **Azure DevOps CI/CD pipelines** that provisioned Windows agents to run parallel automated **Selenium** runs against deployed applications and fed structured results back into work items. Drove a tooling consolidation (**LoadRunner** → **JMeter**) that cut licensing costs while preserving capability. Code review and standards enforcement for the team.

Department of Immigration, Refugees, and Citizenship, Canada

Ottawa, ON

August 2017 – August 2019

Team Lead / Technical Lead

- Led full **Agile / Scrum** delivery — sprint planning, review, retrospective, daily stand-up, backlog refinement, iteration demos, and design walkthroughs; introduced poker-planning estimation and capacity forecasting. **Translated business requirements into working software** through close BA partnership.
- Implemented and deployed **custom web extensions in React / TypeScript / Node.js** with responsive, accessibility-aligned UIs — hands-on while guiding the team. **Mentored developers in SQL Server query optimization** — query tuning, indexing strategy, partitioned tables, and execution-plan analysis against large databases and warehouses.
- Administered departmental **Azure DevOps Server / TFS** instances — installation, configuration, multi-environment upgrades, Kerberos re-configuration — and maintained heavily customized process templates; managed user/group security across multiple project collections.

Department of Justice, Canada Ottawa, ON

December 2015 – August 2017

Development Lead — iCase / Microsoft Dynamics CRM Modernization

- Dev lead on the live **Legal Case Management System (iCase)** — a large **n-tier ASP.NET / C# / ADO.NET** enterprise web application used by **5,000+ lawyers and contracted agents** — while concurrently building its **Microsoft Dynamics CRM** successor. Modified schema and **front-end UI to surface migration status in real time**, and adjusted the integration stored procedures consumed by the department's SAP-based financial system.
- Engineered **Dynamics CRM plug-ins, custom actions, and workflows in C#** (rewriting legacy stored procedures as C# / LINQ); used **SSIS** for ETL between systems; produced **entity-relationship diagrams** and **detailed effort-estimation breakdowns** (FDD / TDD), and applied design patterns (Repository, Strategy, Mediator). **Established code-review processes gated on work items** and mentored junior developers.

Department of Citizenship and Immigration, Canada Ottawa, ON April 2014 – December 2015

Senior Software Engineer · Release Engineering Lead

- Hired alongside one other engineer to take over a release-management environment from a contractor team. **Decommissioned over-engineered legacy C# desktop apps** and replaced them with lightweight **PowerShell calling REST APIs** — most contractor tooling was simply deleted and **everything ended up in source control. Modernization by deletion.**
- Implemented and deployed **web extensions in React / TypeScript / Node.js**; maintained a fork of Microsoft's `vsts-team-calendar` extension with configurable colours and Outlook integration, contributed back upstream. **Installed, configured, customized, and upgraded** the Azure DevOps Server / TFS platform across dev / UAT / production — multi-step upgrades, **Kerberos / SPN configuration**, and database-tier migration.

Department of Justice, Canada Ottawa, ON

October 2008 – April 2014

Full-Stack Developer (2009–2014) · Test-Automation Framework Developer (2008–2009)

Core developer on the department's flagship **Legal Case Management System (iCase)** — a **distributed, multi-tier ASP.NET / C#** enterprise web platform used by **5,000+ government lawyers and contracted agent firms across Canada**, deployed across **load-balanced web servers and regionally distributed document servers** with role- and object-based security, integration to records-management, financial, and email systems, and **24/7 high availability. B2B and B2C** flows: internal lawyers, external agent firms tracking their own billing/timekeeping in iCase, and downstream financial reconciliation.

- Led design and largely solo implementation of the **"My iCase" personalization portal** — the new, customizable widget-based entry point to iCase. Ran JAD sessions, produced prototypes, authored **FDD / TDD documentation**, then built it on **ASP.NET Web Parts with a custom SQL personalization provider, Knockout.js MVVM** (a direct ancestor of modern React patterns), and async AJAX web-service calls. **Evaluated Entity Framework** against the existing Repository-pattern-over-stored-procedures and recommended against adoption on the merits — ORM judgment as much as ORM use.
- **Performance and reliability engineering:** diagnosed a homepage report taking ~10 s uncached — profiled with SQL Profiler, Report Execution logs, and execution plans, then replaced live joins with a nightly-populated flat table (**~50x faster uncached, 3-4x cached**). Built a **self-initiated availability-monitoring suite** across the distributed servers, and diagnosed and resolved a **COM Interop memory leak** on the document tier. Co-led the **SQL Server 2005 → 2012** and **.NET Framework 2.0 → 4.0** upgrades.
- Architected a **Java-based automation framework (IBM Rational Functional Tester)** that **parsed English-language scenario documents into executable scripts with round-trip code generation**, plus a companion manager in **C# / SQL Server** — replacing a labour-intensive **two-week, 20-25-person manual pre-release validation cycle** with automated execution reviewed by one person in five days (**~98% reduction in effort, ~\$160K saved per release cycle**).

Education

Bachelor of Computer Science (Honours) - Carleton University

September 1999 – May 2003

Certifications & Awards

Professional Scrum Master I (Scrum.org) · ITIL v3 Foundation (Loyalist Certification Services) · DOJ Team Merit Award 2012; Team Merit Award 2011; Team Spirit Award 2009

Technical Skills

Languages: TypeScript, JavaScript, C#, SQL (T-SQL), PowerShell, Python, HTML / CSS, XML / XSLT, Go (light)

Front-End / UI: React 19, Next.js, TypeScript, Blazor, Knockout.js MVVM, Vite, JSON, AJAX, jQuery · **responsive design, WCAG / AODA accessibility**

Back-End / APIs: .NET 6 / 8 / .NET Core, ASP.NET / ASP.NET MVC, ADO.NET, Entity Framework, LINQ · Node.js, Express, NestJS · REST API design, GraphQL, WebSocket · Spring Boot (familiarity)

Architecture: n-tier client/server, cloud-native microservices, event-driven workflows, API gateway / management (Azure APIM), design patterns, entity-relationship modelling

Databases: MS SQL Server (2005 → 2019), Azure SQL, PostgreSQL — stored procedures, indexing, partitioning, execution-plan tuning · MongoDB (working familiarity)

Cloud / DevOps: Microsoft Azure (App Service, Functions, SQL, API Management, Key Vault, Storage, Managed Identities, Entra / AAD, Data Factory) · AWS / GCP (working exposure) · Docker, Kubernetes

Source Control / CI/CD: Git, GitHub, GitHub Actions, Azure DevOps Pipelines (YAML) · feature branches, PR-gated CI

Testing: Jest, Vitest, React Testing Library, xUnit / NUnit, Playwright, Selenium

Also experienced with (Microsoft legacy): Visual Studio 2005 → 2022, .NET Framework 2.0 → 4.8, IIS, WCF / WPF, SSRS / SSIS, Crystal Reports, Azure DevOps Server (TFS) administration & upgrades

Process / SDLC: Agile / Scrum (PSM I), Kanban, FDD / TDD with effort estimation, code review, mentoring · Waterfall (regulated environments) · ITIL v3 Foundation